

LECTURE 4.2



What Regression Cannot Do

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

Notes

Weeks already covered are greyed; today's is marked. This is the last one.

Reasons the Estimated Return of 1.52 Might Be Wrong

Since week 6 you have had a single estimate: a return of 1.52 euro an hour for each extra year of schooling.

Notes

Three minutes with your neighbour. Write down every reason this module has given you for why 1.52 might not be the truth. Do not look anything up. In week 6 I asked you to keep a sentence on the least plausible assumption: find it.

Lecture Plan

- **Measurement Error.** Noise in a regressor pulls its coefficient toward zero.
- **Reverse Causality.** Regression is symmetric. Causation is not.
- **The Road to Part II.** One design at a time, named by the problem it solves.

Notes

Reading: [to settle].

The Data-Generating Process Behind the Sample

The objective of these four lines: show what has actually been producing the hundred and twenty rows you have worked with all term.

A_i = ability, drawn once and never recorded

$S_i = 13.5 + \gamma A_i + u_i$

$Y_i = 9.50 + 1.30 (S_i - 12) + \delta A_i + \varepsilon_i$

$S_i^* = S_i + m_i$

and a person appears in your data only if $Y_i \geq f$.

- A_i : ability, which nobody measured
- γ : how much ability raises schooling
- δ : how much ability raises pay
- S_i^* : the schooling the person reported

The Generator as a Causal Diagram

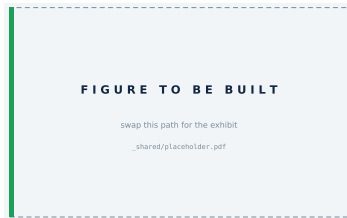


Figure 1: The generator as a causal diagram. Ability points at both schooling and earnings, and it is dashed because nobody measured it. Reported schooling hangs off actual schooling; the wage floor sits on earnings and decides who is in the data at all.

The Estimated Return as the Ability Channel Strengthens

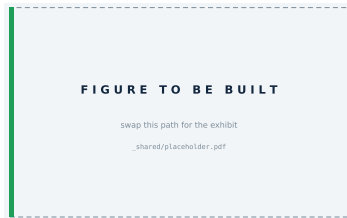


Figure 2: The estimated return, against the strength of the ability channel along the bottom. The truth, 1.30, is the dashed green line. The estimate leaves it as soon as the channel opens.

Nothing in the sample changed. The number moved because ability appears in both equations, and because it is a column nobody has.

Sample Selection Caused by the Wage Floor

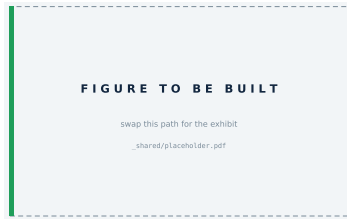


Figure 3: The population in grey with the wage floor drawn across it. Everyone below the line is missing. The fitted line through what survives is flatter than the line through everyone.

This one is not a bias in the estimator. It is a fact about who got into the dataset, and no amount of care with the regression can undo it Heckman (1979).

Why These Faults Leave No Trace in the Residuals

- **4.1** was heteroskedasticity, serial correlation, the wrong shape. Each one leaves a signature in the residuals, and each one has a response.
- **4.2** is ability, mismeasurement and selection. Each one leaves the residual plot looking exactly as it should.

The point of the whole session

There is no diagnostic for today's faults, because a correctly specified regression of the wrong thing on the wrong thing fits beautifully. You cannot test your way out of this. You design your way out, and that is what Part II is for.

SECTION 1

Measurement Error

1. Measurement Error
2. Reverse Causality
3. The Road to Part II

What the Regression Recovers When Schooling Is Mismeasured

The goal of this section: say what noise in a regressor does to its own coefficient, and why it does the opposite of what noise sounds like it should.

The objective of this expression: say what the regression recovers when the x you have is the x you want plus noise.

$$\beta_1 \times \frac{\sigma_S^2}{\sigma_S^2 + \sigma_m^2} \quad (1)$$

- σ_S^2 : the variance of actual schooling
- σ_m^2 : the variance of the reporting error

Why the Reliability Ratio Shrinks the Estimate

The fraction sits between zero and one. So the estimate is smaller than the truth in absolute value, and it is smaller every single time rather than on average. Noisier reporting, smaller coefficient. This is attenuation.

Notes

At zero error the fraction is one and the estimate is the truth. If the reporting error had the same variance as schooling itself, you would recover half the return and your standard error would tell you nothing was wrong.

How the Estimate Falls as Reporting Error Grows

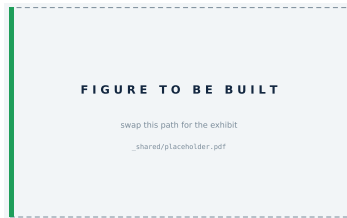


Figure 4: The estimated return against the standard deviation of the reporting error. The dashed green line is the truth. The estimate decays toward zero and never crosses it.

Where the Noise Lands in the Slope Formula

The closed form you derived for homework in week 4 is a covariance over a variance. Adding independent noise to schooling leaves the covariance with earnings alone, because the noise has nothing to do with earnings. It adds to the variance underneath.

This is the counter-intuitive bit

Noise in a regressor does two things, not one. It widens the sampling distribution, which you expected. It also drags the centre of that distribution toward zero, which you did not, and the drag is in the same direction every time.

Three Cases Where Attenuation Does Not Apply

- **Noise in y** does not bias the slope at all. It inflates the standard error and leaves the coefficient where it was.
- **Noise in a control** in a multiple regression can push the coefficient you care about in either direction.
- **Non-classical error** breaks the result outright: people who round their schooling up are not a random selection Bound et al. (2001).

Notes

The classical case is the one with a clean answer, and the literature on the return to schooling has argued about exactly this since Griliches (1977). It is also why the twin studies are awkward: differencing two siblings removes the family, but it removes far more real schooling than reporting error, so the differenced estimate is the more attenuated one Ashenfelter & Krueger (1994).

Vote: Which Survey Gives the Smaller Estimate?

Two surveys measure the same return. In one, people report their own years of schooling. In the other, the years come from the school register.

Notes

Vote, argue with your neighbour, vote again. Then: is the difference between them a matter of luck, or would it come out the same way again?

Pause

Notes

Two minutes. Name a variable in something you have studied that is certainly measured with error, and say which way you now think its coefficient is wrong.

SECTION 2

Reverse Causality

1. Measurement Error
2. Reverse Causality
3. The Road to Part II

The Same Data Fitted in Both Directions

The goal of this section: show that the regression is symmetric in its two variables, that causation is not, and that the output cannot tell them apart.

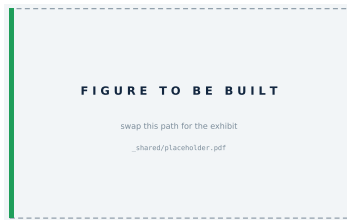


Figure 5: The same hundred and twenty people, with earnings on schooling in navy and schooling on earnings in green. Both are least squares. Both are correct. They are different lines.

The Algebra Behind the Two Fitted Slopes

The objective of these two expressions: show that the difference between the two fits is arithmetic rather than rhetorical.

$$\hat{\beta}_{Y|S} = \frac{s_{SY}}{s_S^2} \qquad \hat{\beta}_{S|Y} = \frac{s_{SY}}{s_Y^2} \qquad (2)$$

- $\hat{\beta}_{Y|S}$: the slope from regressing earnings on schooling
- $\hat{\beta}_{S|Y}$: the slope from regressing schooling on earnings
- s_{SY} : the sample covariance of schooling and earnings
- s_S^2 : the sample variance of schooling
- s_Y^2 : the sample variance of earnings

The numerator is the same in both. The two slopes differ only in which variance sits underneath, and their product is the R-squared you learned to read in week 5.

Simultaneity: When Causation Runs in Both Directions

Sometimes the honest picture is not one direction or the other but both. Price and quantity are settled together, so a regression of quantity on price traces out neither the demand curve nor the supply curve Working (1927). Naming that problem is what founded the subject Haavelmo (1943).

And in our own example

If you stay in school longer because you expect to earn more, then earnings expectations cause schooling at the same time as schooling causes earnings. Both arrows are live, and the slope is a blend of the two.

The Regression Output Is Identical Either Way

- The R-squared is identical whichever variable you called y .
- The t-statistic is identical.
- The residual plot is clean in both directions.
- A larger sample makes both answers more precise and neither more correct.

Direction is not in the data. It comes from timing, from institutions, from what could not possibly have caused what - and, in Part II, from a design built to make one direction the only one available.

Vote: Do More Police Cause More Crime?

Areas with more police officers per head have more recorded crime.

Notes

Vote, argue with your neighbour, vote again. Then: what evidence would settle it, and could any regression on these two columns ever produce that evidence?

Pause

Notes

Two minutes. Write down a published claim you have met this year where you now suspect the arrow runs the other way. Keep it for the next slide but one.

SECTION 3

The Road to Part II

1. Measurement Error
2. Reverse Causality
3. The Road to Part II

Instrumental Variables, for a Confounder Nobody Measured

The goal of this section: name each remaining problem and the design that exists to answer it. One idea each, and no technique.

The idea

Find something that moves schooling and has no other route to earnings. Use only the part of schooling that it moves, and throw the rest away.

Compulsory-schooling laws move how long some people stay in school without having anything else to do with their wages Angrist & Krueger (1991). The schooling literature is largely a long argument about which of these candidates survives scrutiny Card (1999).

Difference-in-Differences and Fixed Effects, for What Is Constant

The idea

If the thing you cannot see does not change, compare a unit with itself over time, or compare the change in a treated group against the change in an untreated one. Whatever is constant subtracts out.

Employment in fast-food restaurants either side of a state border, before and after one state raised its minimum wage Card & Krueger (1994). Or two siblings with the same parents and different schooling Ashenfelter & Krueger (1994).

Notes

What is differenced away never has to be measured. That is the whole trick, and it is also the whole limitation: anything that changes is still there.

Regression Discontinuity, for a Rule That Assigns Treatment

The idea

When a published threshold decides who is treated, the people just above and just below it are otherwise alike. The jump at the cutoff is the effect.

The design was invented to evaluate scholarship awards handed out on a test score Thistlethwaite & Campbell (1960), and economics found it again in a class-size rule that splits a year group once it passes forty pupils Angrist & Lavy (1999). The survey to start from is Lee & Lemieux (2010).

What the Three Designs Share and What They Cost

- Each one buys a comparison you can defend out loud.
- Each one pays for it with a narrower question and fewer people it applies to.
- None of them is a better estimator. Least squares is usually still what runs.

The change is in where the credibility comes from. It stops coming from the model and starts coming from how the data were generated. That argument was made against the regressions of the day by Leamer (1983), taken up as a programme by Angrist & Pischke (2010), and given a Nobel lecture by Card (2022).

Why These Designs Are Left Until Part II

Angrist & Pischke (2017) argue that a first course built on assumptions and corrections leaves students unable to read a modern applied paper, and the survey evidence is that most first courses still look much like that Harter & Asarta (2022). Against them: Sims (2010), that an approach which answers only the questions an experiment can reach leaves most of economics unanswered; Keane (2010), that a design rests on assumptions too, only unstated ones; and Deaton (2010), that an estimate you can defend here is not one you can carry elsewhere.

Where this module stands

You cannot judge an instrument without knowing what a standard error is, and you cannot see why a design is needed until you have watched a regression fail. Part I was that, deliberately. Part II is where the designs are taught.

Vote: Which Design Would You Reach For?

A maintenance grant is paid to every student whose Leaving Certificate points fall below a published cut-off. Does the grant raise graduation rates?

Notes

Vote between the three, argue with your neighbour, vote again. Then: what would have to be true about students either side of the cut-off?

Returning to the Opening Question From Week 1

[The hook from 1.1, restated word for word as it was put in week 1.]

Notes

You split on it in September and could not say why. Say why now, in one sentence each, using this term's vocabulary.

What Would Have To Be True?

The standing question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

[Candidate this week: one published estimate, circulated in advance. Groups of four, seven minutes, four questions.]

1. What is the number, in its units, and what does the sentence around it claim?
2. What would have to be true of the population for that sentence to be right?
3. Which of this term's faults is the most plausible story where it isn't?
4. What design would answer it, and what would that design cost you?

Application

Open the generator and turn the switches on one at a time.

Notes

Vote first, before anything runs: if I dial the reporting error up, does the estimate get noisier, get smaller, or both? Take the vote, then run it.

Then the second run, and it is the one that matters. Turn ability on, refit, and put last week's diagnostics on the result. Every test passes.

What Regression Cannot Do

Today was three separate ideas, each doing one job:

1 Measurement Error.

Noise in a regressor pulls its coefficient toward zero, and it does so every time rather than on average.

2 Reverse Causality.

Regression is symmetric and causation is not, and no line of the output records which direction you meant.

3 The Road to Part II.

Each design exists because one of these faults cannot be diagnosed, only designed around.

Notes

Part II begins where this stopped: not with a better estimator, but with a better reason to believe the comparison.

What to Read and Do Before Part II

- **Read** Angrist & Pischke (2017) on what a first course owes you, then one of Sims (2010), Keane (2010) or Hansen (2017) on what it would cost to give you only that. Two hundred words on which you find more persuasive, and why.
- **Do** the appendix derivation. It is the one result from today that you can get to in four lines from the slope formula in week 4.
- **Try** the app with every switch on at once, and see whether any diagnostic you know would have caught it.

Problem set 4

[Out this week, due week n. Covers 4.1 and 4.2, and the last question is the standing question asked of a paper you choose yourself.]

Thank you

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BACKUP

Appendix

Derivation: Attenuation From Classical Measurement Error

Write the slope as a covariance over a variance, substitute reported schooling for actual schooling, and see which of the two pieces the noise touches.

Notes

Three steps. Substitute $S^* = S + m$ into the numerator and use the assumption that m is independent of everything else, so the numerator does not change. Substitute it into the denominator, where the variances add. Divide. The step that does the work is the second, and the whole result is that the noise lands underneath and not on top.

[Full derivation. Set as homework; worked solution published with the set.]

Notes and Tools

- `_apps/regression/R/model.R`, block B_04: the switches, and what each one does.
- The measurement-error app, and the script that sweeps the reporting error.
- `lm(y ~ x)` and `lm(x ~ y)` on the same data, which is a two-line demonstration.
- The reading list for Part II, on the module page.

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